

Chenjie Qiu

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Profile

My research focuses on reliable and interpretable LLM systems connecting external evidence, agent behavior, internal representations, and user state. I study how models and agents organize evidence, make decisions, diagnose failures, and determine when self-correction is justified. My work spans structure-aware retrieval and reasoning, verifiable language agents, and mechanistic interpretability, using counterfactual evaluation, probing, and activation interventions to identify information that causally shapes behavior.

Education

Guangdong University of Technology

M.S. in Software Engineering

Advisors: Bingfeng Chen and Boyan Xu

Guangzhou, China

Sep 2023 – Jun 2026

Guangdong University of Finance

Bachelor, E-Commerce

Guangzhou, China

Sep 2018 – Jun 2022

Research Interests

Structured Evidence and Verifiable Language Agents: Developing embedding, retrieval, planning, and verification methods for language agents that organize and reason over structured evidence such as graphs, tables, schemas, and interaction traces. I study verifiable self-correction: attributing failures, calibrating uncertainty, and deciding when to repair, verify, or abstain.

Mechanistic Interpretability of Model and User States: Using probes, representation analyses, and activation interventions to study how language models encode and use retrieved evidence, affective signals, and user-state information across reasoning and multi-turn dialogue.

Research Papers

Primary Research Contributions

1. B. Chen, **C. Qiu**, Y. Xie, B. Xu, R. Cai, and Z. Hao. “ $\mathcal{S}^2$ IT: Stepwise Syntax Integration Tuning for Large Language Models in Aspect Sentiment Quad Prediction.” *Findings of NAACL 2025*.
Addresses the difficulty of integrating global and local syntactic structure in LLM-based aspect sentiment quad prediction through a stepwise syntax-integration tuning framework.
2. **C. Qiu**, Y. Luo, B. Chen, R. Cai, Z. Hao, and B. Xu. “GraphPack: Query-Aware Subgraph Packing for Graph-Enhanced Large Language Models.” Submitted to *EMNLP 2026*.
Addresses the challenge of selecting compact and structurally coherent graph evidence under fixed context budgets through query-aware subgraph packing and query-conditioned graph encoding.
3. **C. Qiu**, B. Chen, S. Shi, and B. Xu. “RippleRAG: Learning Which Evidence to Propagate across Retrieval Goals for Multi-Hop Retrieval-Augmented Generation.” Submitted to *AAAI 2027*.
Addresses the difficulty of valuing evidence by its downstream utility across dependent multi-hop retrieval goals through Goal-DAG-based evidence propagation and a reachability-informed critic.

Co-Authored Research Contributions

1. R. Cai, M. Lu, **C. Qiu**, Z. Hao, and B. Xu. “Directional Dispersion Trees for Incremental Long-Document RAG.” Submitted to *ACL ARR 2026 August*.
2. Z. Zeng, **C. Qiu**, Y. Wang, Z. Li, R. Cai, and B. Xu. “RoTA: Role-Preserving Table Alignment for Frozen LLMs.” Submitted to *ACL ARR 2026 August*.
3. R. Cai, J. Li, **C. Qiu**, Z. Hao, and B. Xu. “GeoRAG: Geometric Indexing and Adaptive Reasoning for Retrieval-Augmented Generation.” Submitted to *AAAI 2027*.
4. S. Shi, Q. Xu, W. Yang, **C. Qiu**, B. Chen, and B. Xu. “EvoEM: Evolutionary Embedding Model for Long-term Interactive Schema Linking.” Submitted to *AAAI 2027*.
5. S. Shi, Q. Xu, W. Yang, **C. Qiu**, B. Chen, and B. Xu. “ReliORLM: Internalizing Step-Level Disambiguation for Ambiguous Natural Language to Operations Research Modeling.” Submitted to *AAAI 2027*.

Research Experience

DMIR Lab, School of Computer Science and Technology

Guangzhou, China

Structured Evidence Retrieval and Multi-Hop Reasoning

Feb 2025 – Present

- Led the design of **GraphPack**, reframing query-specific subgraph retrieval as budgeted evidence allocation that jointly optimizes semantic relevance and structural compactness under a fixed context limit, with query-conditioned encoding to preserve the topology of packed evidence.
- Improved F1 over the strongest reported baselines by 6.90 points on WebQSP and 5.97 points on CWQ.
- Led the design of **RippleRAG**, formulating multi-hop retrieval as execution over a question-specific Goal-DAG in which each goal conditions on evidence inherited from its dependencies, and trained a reachability-informed critic with counterfactual write-no-write look-ahead to predict whether a passage write improves downstream retrieval.
- Achieved state-of-the-art retrieval performance under matched corpus-level evaluation, ranking first in all twelve dataset-metric comparisons across MuSiQue, HotpotQA, and 2WikiMultiHopQA; matched no-critic ablations show that the Reachability Critic contributes 3.14 Recall@5 points on MuSiQue and 1.51 points on 2WikiMultiHopQA.

Affective and User-State Representations in Multi-Turn Dialogue

2026 – Present

- Investigated how affective user state is represented and carried across dialogue turns using linear and speaker-conditioned probes over residual-stream activations.
- Designed lexical, current-turn, and history-controlled evaluations to separate surface-word cues from history-dependent latent traces.
- Used linear and speaker-conditioned probes to identify affective and user-state representations, and conducted activation-patching and steering checks to examine whether these representations are reflected in subsequent generation.

Structure-Aware Sentiment Reasoning

Sep 2023 – Sep 2024

- Led the design of **$\mathcal{S}^2\text{IT}$** (Findings of NAACL 2025), a stepwise syntax-integration framework that decomposes aspect-sentiment quad prediction into global syntax-guided extraction, local syntax-guided classification, and fine-grained structural tuning.
- Achieved state-of-the-art F1 across five supervised benchmarks, including a 2.71-point gain on Restaurant-ACOS; ablations show that removing syntax information lowers F1 by 1.3 points on Restaurant and 1.1 points on Laptop in the 7B setting, with further drops when structural tuning tasks are removed.

Patents

- **Method and System for Sentiment Analysis Based on Structure-Aware Language Models.** Chinese invention patent application No. 202510140392.X, entered substantive examination, Jul 2025.
- **A Text-Graph Subgraph-Enhanced Method for Graph Question Answering.** Chinese invention patent application No. 202510648762.0, entered substantive examination, Aug 2025.

Awards

- Graduate Entrance Scholarship, Guangdong University of Technology 2023
- Third-Class Academic Scholarship, School of Computer Science and Technology 2024
- Second-Class Academic Scholarship, School of Computer Science and Technology 2025

Skills

- **Research Methods:** Retrieval-augmented generation, structured evidence retrieval, linear probing, activation patching, and activation steering.
- **LLM Development:** Model fine-tuning, inference, and evaluation with PyTorch, Hugging Face Transformers, and PEFT.
- **Agent Workflows:** Building agent workflows with LangChain and LangGraph.
- **Research Infrastructure:** Managing research servers and Linux GPU environments; Conda/CUDA, Docker, and remote troubleshooting.
- **Languages:** Mandarin, Cantonese, and English (CET-6; IELTS in preparation).